

International Student Mobility and Economic Contribution: Australia's Position in the Global Education Market

FIT5147 Data Exploration and Visualisation

Monash University

DEP Part 2: Project Report

Student Name: Enkhjargal Unurjargal

Student ID: 35314516

Teaching Associate: Shruti Sandeep Mahajan (Applied 03)

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1. Introduction

International education is a critical component of Australia's service economy, contributing significantly to export revenue while also shaping labour markets, housing demand, and broader economic activity. At the same time, increasing global competition from destinations such as the United Kingdom, Canada, and the United States has intensified the importance of understanding international student mobility as both an economic and strategic issue.

This report investigates Australia's position within the global education market through a multi-dimensional data exploration approach. The analysis integrates sector-level enrolment data, source-country composition, export revenue indicators, and global mobility metrics to provide a comprehensive view of how international education evolves over time and across contexts.

The study is guided by three key research questions:

1. What long-term patterns and structural changes can be observed in international student enrolments across sectors and source countries?
2. How are international student enrolments associated with Australia's education export performance?
3. How does Australia compare with other major study destinations in terms of international student mobility?

By addressing these questions, the analysis moves beyond simple trend description and instead evaluates structural dynamics, economic relationships, and global positioning within the international education system.

2. Data Wrangling and Checking

2.1 Data Sources and Analytical Rationale

This study integrates five complementary datasets to capture different dimensions of international education:

1. Sector enrolment dataset (Australian Government Department of Education, 2025a) — provides longitudinal data on enrolments and commencements from 2005 to 2025, enabling time-series and structural analysis. Where 2025 data were not yet finalised at the time of analysis, the year-to-date December 2024 release (Australian Government Department of Education, 2024) was used as a supplementary source to confirm earlier observations. Monthly release schedules were also consulted to confirm data currency (Australian Government Department of Education, 2025b).
2. Source-country datasets (Australian Government Department of Education, 2024 and 2025c) — capture geographic composition at two time points (2015 and 2025), allowing analysis of market concentration and directional change. These were accessed via the country-level summary tables rather than the sector-level files.
3. Education export dataset (Australian Bureau of Statistics, 2024 and 2025) — measures economic contribution in both calendar-year and financial-year formats. The calendar-year release (covering to 2024) was used as the primary source for revenue analysis, with the financial-year release (2024–25) used to verify the most recent period. Direction-of-trade data

from the Department of Foreign Affairs and Trade (2025) was also consulted to cross-check service export values against an independent source.

4. Global mobility dataset (Our World in Data, 2025) — provides internationally comparable indicators for cross-country benchmarking. Note: this dataset was substituted from the proposed OECD source due to superior temporal coverage and accessibility, all findings remain consistent with the original analytical intent.

The use of multiple dataset versions across agencies reflects the reality that different sources are updated on different schedules, and that no single file covers the full analytical scope of this study. Where datasets overlapped in coverage, values were cross-checked for consistency before use.

2.2 Data Wrangling Process

Substantial transformation was required to ensure comparability across datasets with different structures and reporting conventions. The sector enrolment dataset was originally provided in wide format, with years spread across columns. It was reshaped into long format so that each row represented a single sector–year observation. This transformation is critical for enabling consistent time-series analysis and flexible visualisation using `ggplot2`, particularly when comparing multiple categories across time. The dataset was then filtered to four primary sectors (Higher Education, VET, ELICOS, and Schools), improving interpretability while retaining the dominant structure of Australia's international education system.

The source-country data were provided as two separate .xls snapshot files (2015 and 2025), loaded using `readxl`. After loading, variables were standardised and total student counts constructed by combining relevant components. The two datasets were then merged into a unified structure to support direct two-point comparison. Aggregate category rows were removed to prevent double-counting and ensure valid comparisons.

The education export dataset was reported in financial years (e.g., 2024–25), while enrolments used calendar years. To align these, the starting year of each financial period was used as a proxy (e.g., 2024–25 mapped to 2024). This approach maintains temporal consistency while avoiding unnecessary complexity that would not materially affect annual comparisons.

Finally, the global mobility dataset required harmonisation of country names and filtering for the ten selected destination countries, ensuring accurate joins and reliable geographic representation in the choropleth map.

2.3 Data Checking and Validation

Multiple validation steps were implemented to ensure data integrity:

1. Type validation ensured that all numerical variables were correctly parsed and usable for analysis.
2. Missing data handling removed incomplete observations where they would distort interpretation.
3. Category standardisation ensured consistency in sector and country naming across datasets, including resolving variations such as "Republic of Korea" and "Korea, Republic of" to a single label.
4. Temporal consistency checks verified alignment between datasets with different reporting structures, including the financial-to-calendar year conversion described above.

In addition to these technical checks, visual validation was used to confirm plausibility. Observed patterns — such as sustained growth prior to COVID-19 and sharp disruption during 2020–2021 — aligned with real-world expectations, providing additional confidence in data quality.

2.4 Tools used

All data preparation, analysis, and visualisation were conducted in R. The workflow utilised:

1. *tidyverse* for data transformation and manipulation
2. *ggplot2* for visualisation
3. *readxl* for structured data extraction from Excel and .xls files across all datasets
4. *forecast* and *broom* for time-series modelling and statistical analysis

R was selected due to its ability to integrate the full analytical pipeline within a single environment, ensuring reproducibility and consistency across all stages of the analysis.

3. Data Exploration

This section examines the data through a sequence of complementary visual analyses. Each visualisation is selected based on the structure of the data and the specific analytical objective, ensuring that insights are not only descriptive but also methodologically justified.

3.1 Sector Level Enrolment Dynamics

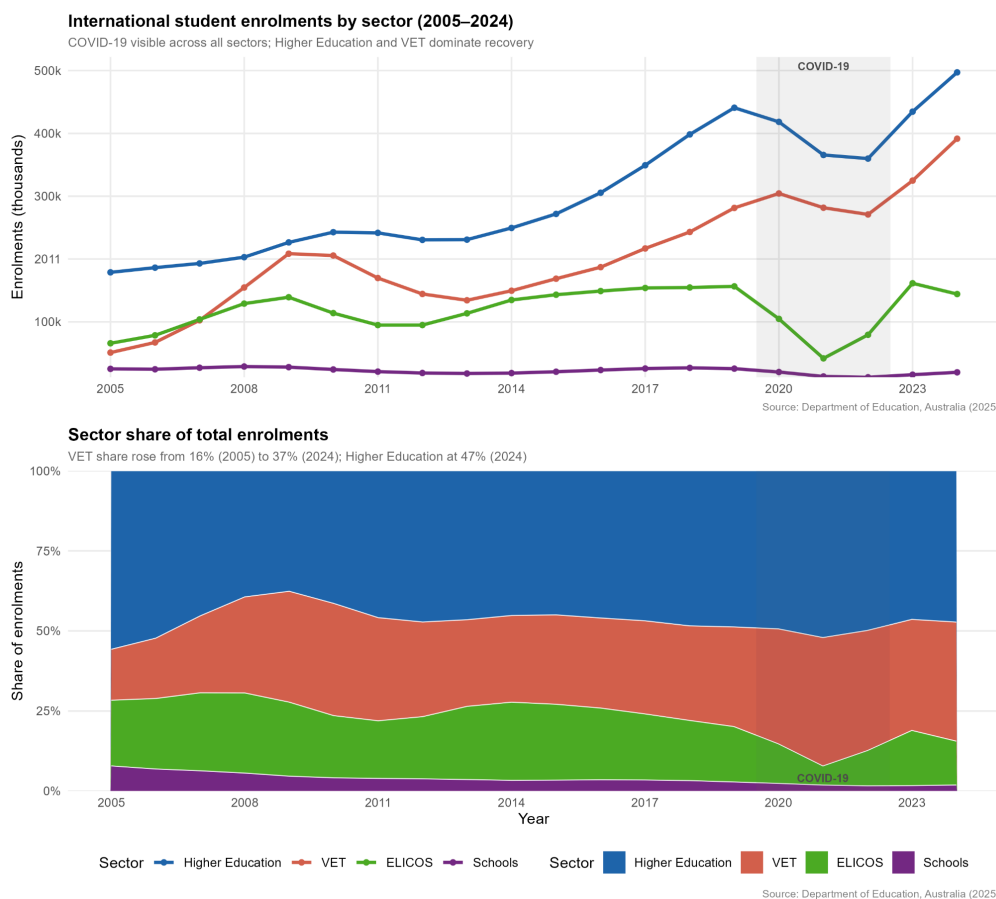


Figure 1

International Student Enrolments by Sector and Sector Share of Total Enrolments, Australia, 2005–2024

Note. The top panel shows absolute enrolments (thousands) by sector using a multi-series line chart. The bottom panel shows proportional sector composition using a stacked area chart. Source: Australian Government Department of Education (2025a).

The multi-series line chart captures longitudinal patterns in enrolments across sectors. This representation is effective for continuous time-based data, as it preserves temporal ordering while enabling direct comparison between categories. It also makes structural turning points - such as the COVID-19 disruption - visually explicit (Australian Government Department of Education, 2025a).

The results reveal sustained pre-pandemic growth in Higher Education and VET, followed by a sharp contraction and uneven recovery. The differing recovery trajectories indicate that external shocks do not affect all sectors equally, implying that institutional resilience and policy responses must be tailored rather than uniform.

The proportional stacked area chart complements this perspective by shifting focus from absolute magnitude to relative composition. By normalising totals, it highlights how sector shares evolve over time. The analysis shows that Australia's international education system is not only expanding in volume but also undergoing structural rebalancing, with implications for long-term market positioning and strategic planning.

3.2 COVID-19 Disruption and Structural Differences

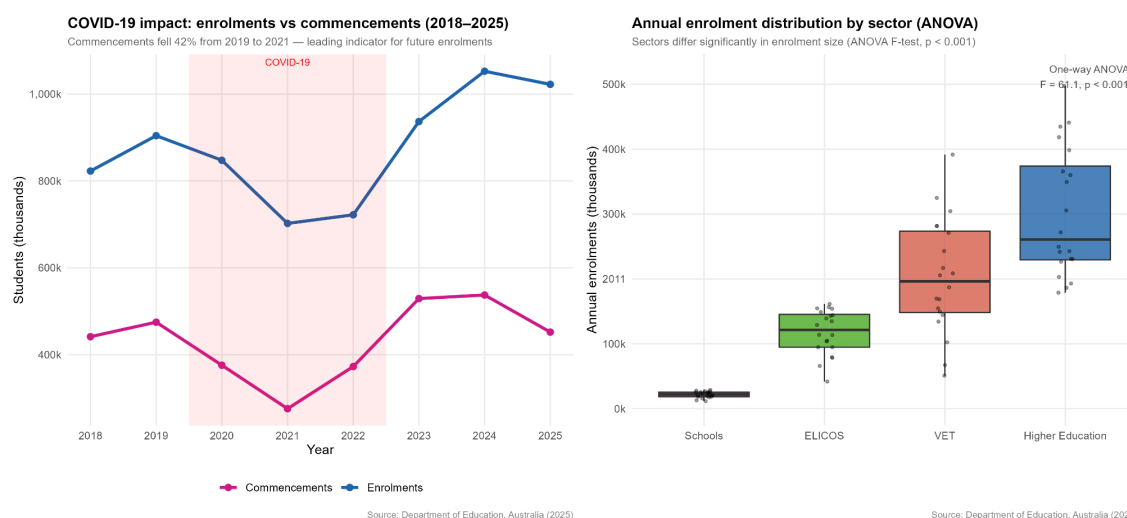


Figure 2

COVID-19 Impact on Enrolments and Commencements, and Annual Enrolment Distribution by Sector, 2018–2025

Note. The left panel contrasts total enrolments with commencements as a leading demand indicator. The right panel displays enrolment distributions across sectors with one-way ANOVA results ($F > 1$, $p < 0.001$). Post-hoc Tukey HSD tests confirmed all pairwise sector differences were statistically significant ($p < 0.001$). Source: Australian Government Department of Education (2025a).

The dual-line chart contrasts enrolments with commencements, distinguishing between total student stock and new inflows. This distinction is analytically important, as commencements serve as a leading indicator of demand. The sharper decline in commencements during 2020–2021 demonstrates that border restrictions primarily disrupted new student inflows rather than the existing population.

The boxplot, supported by ANOVA testing, provides a structural comparison of enrolment distributions across sectors. Unlike time-series visualisations, this approach summarises central tendency and variability, allowing meaningful comparison of sector scale. The statistically significant ANOVA result confirms that observed differences are systematic rather than random.

Together, these approaches reveal both short-term disruption dynamics and persistent structural inequalities between sectors. Post-hoc Tukey HSD tests confirmed that all pairwise sector differences were statistically significant ($p < 0.001$), establishing that Higher Education, VET, ELICOS, and Schools each operate at structurally distinct enrolment scales rather than representing incidental variation. This suggests that recovery trajectories are shaped not only by external shocks but also by inherent differences in sector size and composition, with implications for sector-specific policy responses.

3.3 Source-country composition and change

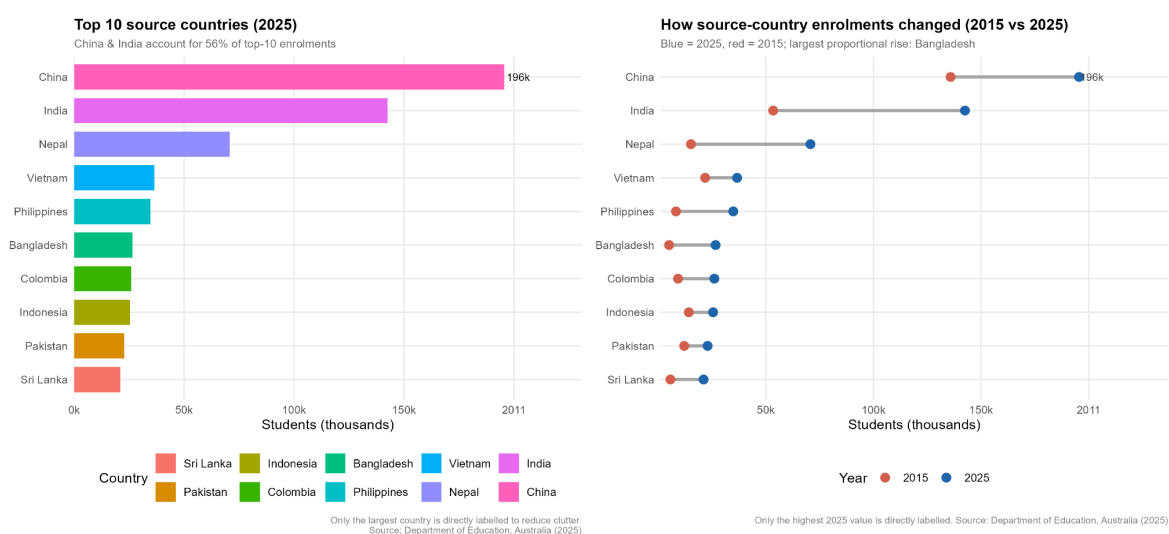


Figure 3

Top 10 Source Countries by Enrolment Volume (2025) and Change in Source-Country Enrolments (2015 vs. 2025)

Note. The left panel ranks source countries by total enrolments in 2025. The right panel shows the two-point comparison of enrolment levels between 2015 and 2025 for the same countries. Source: Australian Government Department of Education (2024, 2025c).

The horizontal bar chart enables clear ranking of source countries and is well suited for categorical comparison, particularly where labels are lengthy. The results demonstrate a high degree of concentration, with a small number of countries accounting for a large proportion of enrolments. This indicates structural dependence on key markets.

The two-point comparison chart is used instead of a full time series due to data limitations. This avoids implying false continuity while still communicating direction and magnitude of change between two comparable periods.

The findings show that while dominant countries such as China and India remain influential, the composition of source markets is evolving. This highlights a critical policy implication: heavy reliance on a limited number of countries increases vulnerability to geopolitical and economic disruptions, reinforcing the importance of diversification strategies.

3.4 Economic Contribution and Revenue Dynamics

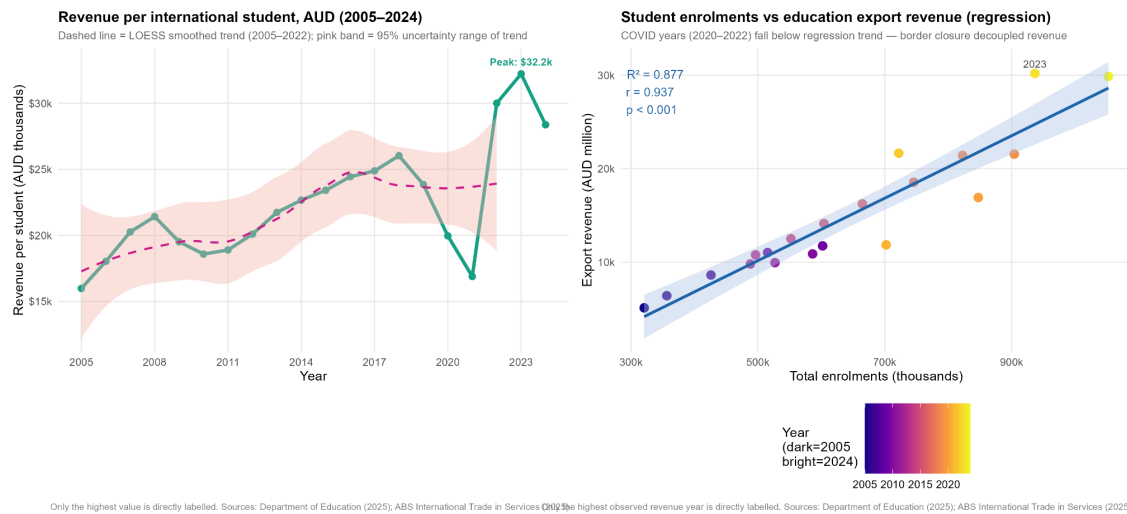


Figure 4

Revenue per International Student in AUD (2005–2024) and Relationship Between Total Enrolments and Education Export Revenue

Note. The left panel shows revenue per student with LOESS smoothed trend line and 95% uncertainty band. The right panel displays a scatterplot with linear regression ($R^2 = 0.877$, $r = 0.937$, $p < 0.001$), COVID-19 years (2020–2022) fall below the regression trend. Sources: Australian Government Department of Education (2025a); Australian Bureau of Statistics (2024).

The revenue-per-student line chart examines changes in economic intensity over time. The inclusion of a LOESS smoother allows underlying trends to emerge without imposing a rigid functional structure. The upward trend in recent years suggests that economic value per student is increasing, indicating a shift beyond purely volume-driven growth (Australian Bureau of Statistics, 2024; Australian Government Department of Education, 2025a).

The scatterplot with regression evaluates the relationship between enrolments and export revenue. This method is appropriate for analysing associations between continuous variables and provides a direct visual measure of correlation. The strong positive relationship confirms that international education is a major contributor to Australia's export economy.

However, deviations observed during the COVID-19 period reveal a temporary decoupling between enrolments and revenue. This suggests that external shocks can disrupt established economic relationships, implying that reliance on student volume alone may not fully capture the sector's economic dynamics.

3.5 Long-Run Trends and Structural Deviations

The trend comparison chart introduces a benchmark trajectory, allowing observed enrolments to be interpreted relative to expected growth. This approach is designed for contextual analysis rather than forecasting accuracy.

The diverging bar chart visualises deviations from the trend, clearly distinguishing between above- and below-trend periods. This format is particularly effective for highlighting magnitude and direction simultaneously.

The analysis shows that COVID-19 produced the largest negative deviation in the observed period, followed by a rapid recovery. This quantifies the scale of disruption and demonstrates that the sector's rebound exceeded historical expectations, suggesting strong underlying demand resilience.

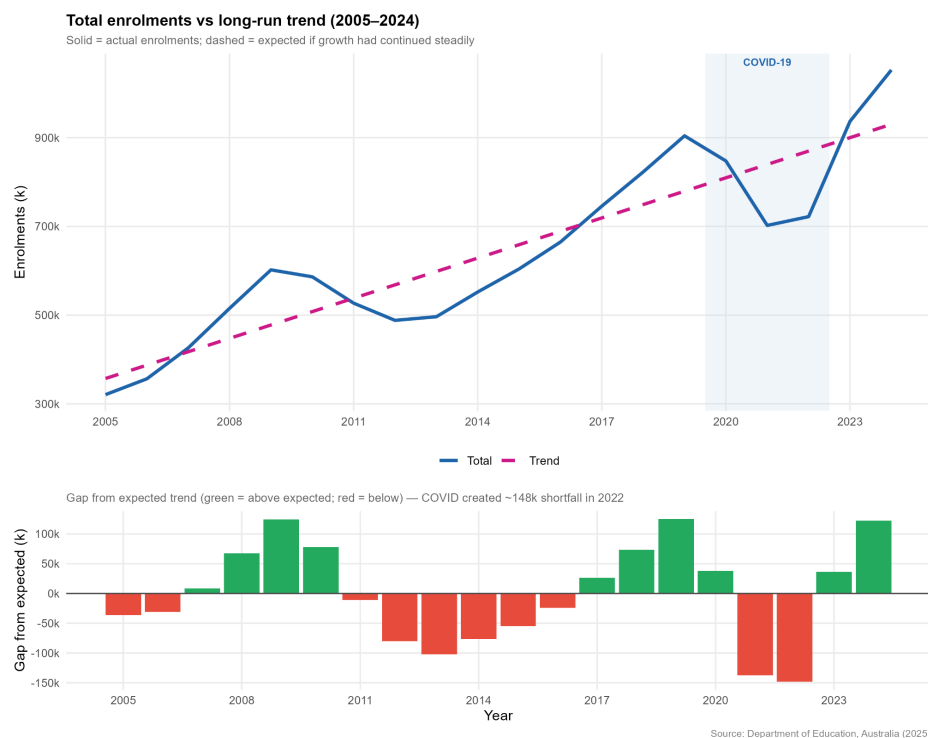


Figure 5

Total International Student Enrolments Compared to Long-Run Trend, and Annual Deviation from Expected Trend, 2005–2024

Note. Top panel compares actual enrolments (solid line) against expected steady-growth trajectory (dashed line). Bottom panel shows annual gap from trend, green bars indicate above-trend years, red bars indicate below-trend years. COVID-19 produced an estimated shortfall of approximately 148,000 students in 2022. Source: Australian Government Department of Education (2025a).

3.6 Forecasting as Supplementary Insight

The forecast visual integrates historical data, projected values, and confidence intervals, providing a transparent representation of uncertainty. This is essential for responsible interpretation of predictive models.

The projections indicate continued recovery, although the wide confidence intervals highlight substantial uncertainty. This reinforces that forecasts should be interpreted as indicative scenarios rather than precise predictions.

The inclusion of forecasting serves to demonstrate methodological capability while maintaining analytical caution, ensuring that the report does not overstate predictive conclusions.

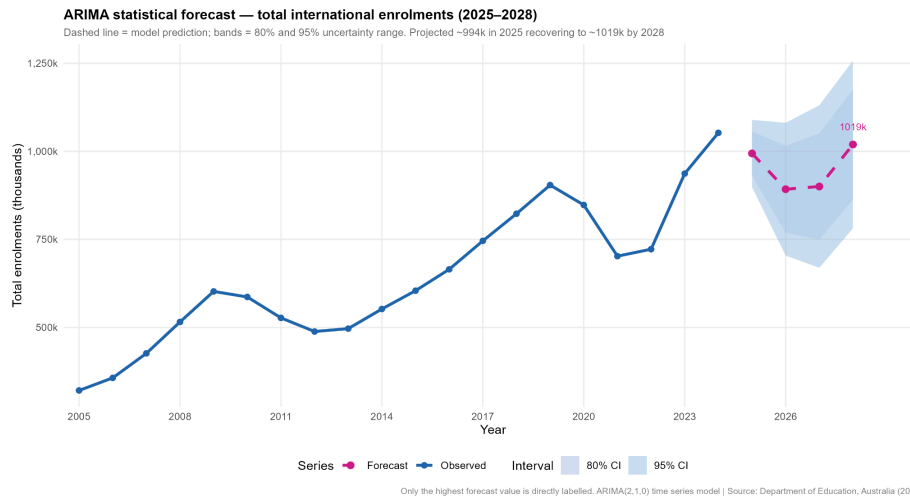


Figure 6

ARIMA Statistical Forecast of Total International Student Enrolments, 2025–2028

Note. Dashed line represents model predictions from an ARIMA(2,1,0) time series model. Shaded bands indicate 80% and 95% confidence intervals. Projected enrolments recover from approximately 994,000 in 2025 to approximately 1,019,000 by 2028. Source: Australian Government Department of Education (2025a).

3.7 Australia's Global Position

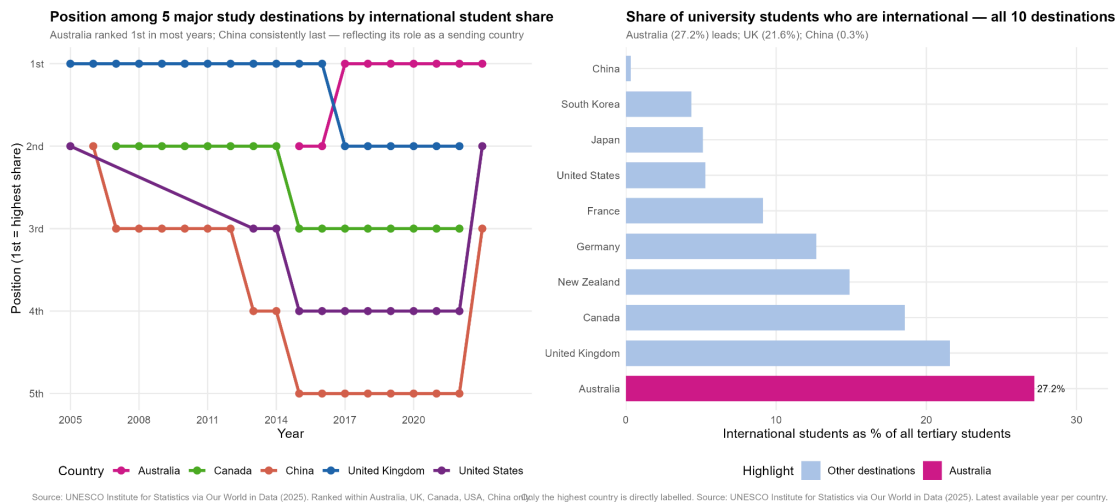


Figure 7

Australia's Position Among Five Major Study Destinations by International Student Share (2005–2022) and Share of University Students Who Are International Across Ten Destinations (Latest Available Year)

Note. Left panel shows longitudinal ranking among Australia, Canada, China, United Kingdom, and United States. Right panel displays cross-sectional comparison of internationalisation rates, Australia leads at 27.2%. Source: Our World in Data (2025), derived from UNESCO Institute for Statistics data. Latest available year used per country.

The ranking line chart focuses on relative position rather than absolute scale, making it appropriate for analysing competitive standing. The results show that Australia consistently ranks among the top international education destinations.

The cross-sectional bar chart provides a broader comparison using a normalised metric, enabling meaningful comparison across countries of different sizes (Our World in Data, 2025).

Together, these visualisations confirm that Australia is both historically and currently a highly internationalised education destination. This positioning reflects a strong competitive advantage, but also implies dependence on maintaining global attractiveness.

3.8 Global context

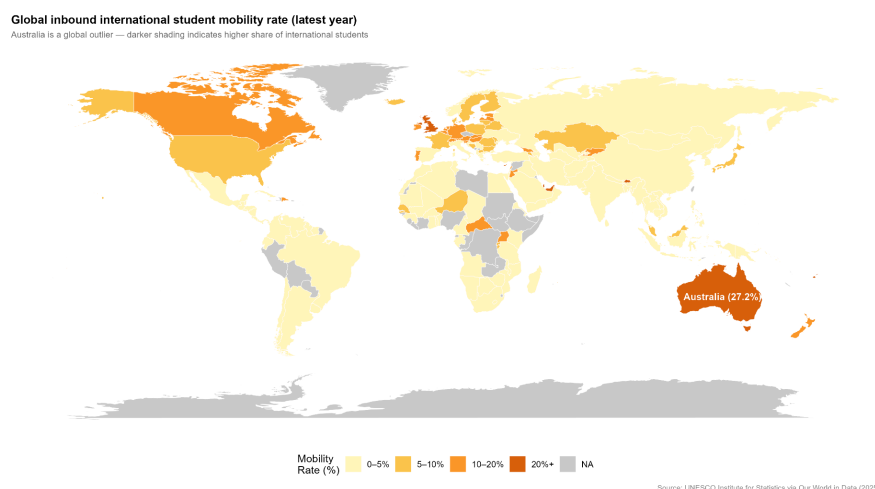


Figure 8

Global Inbound International Student Mobility Rate by Country (Latest Available Year)

Note. Choropleth map displays binned international student mobility rates. Darker shading indicates a higher share of international students among total tertiary enrolments. Australia (27.2%) is highlighted as a global outlier. Grey shading indicates data not available. Source: Our World in Data (2025), derived from UNESCO Institute for Statistics data.

The choropleth map illustrates global variation in international student mobility rates. This visual form is suited to spatial analysis, enabling patterns of geographic concentration to be identified, even though it is less precise for exact comparison.

The map shows that high levels of internationalisation are concentrated in a small number of countries, with Australia clearly positioned as a global outlier. This reinforces that Australia's education system operates at an unusually high level of international engagement, distinguishing it from most countries.

This spatial perspective complements earlier findings by demonstrating that Australia's position is not only strong in absolute terms, but also exceptional in a global context, highlighting both its competitiveness and its structural reliance on international students.

3.9 Overall interpretation of the exploration process

Across all analyses, visualisation methods were selected to align with both data structure and analytical purpose. Time-series charts captured temporal dynamics, bar charts enabled categorical

comparison, boxplots supported statistical evaluation, scatterplots revealed relationships, and maps provided spatial context.

The visual patterns observed in Figures 7 and 8 consistently position Australia as a global outlier in international student mobility. The ranking chart confirms Australia held the top position among major destinations across most years, while the cross-sectional comparison shows its 27.2% international student share substantially exceeds all other destinations in the comparison group. Together, these findings confirm that Australia's internationalisation level is not only strong in absolute terms but structurally distinct from other major study destinations, reinforcing both its competitive advantage and its dependence on maintaining global attractiveness.

4. Conclusion

In response to Q1, the analysis revealed sustained sectoral growth from 2005 interrupted by COVID-19, with Higher Education and VET dominating recovery while the structural composition of enrolments continued to shift — notably VET's share rising from 16% to 37% by 2024. In response to Q2, a strong positive relationship ($R^2 = 0.877$) between total enrolments and education export revenue was confirmed, with COVID-19 producing a temporary decoupling that highlights the sector's vulnerability to external shocks. In response to Q3, Australia consistently ranked first among major study destinations with an international student share of 27.2%, a position reinforced by both longitudinal ranking analysis and cross-sectional comparison with nine other destination countries.

At the same time, the findings confirm a strong positive relationship between student enrolments and export revenue, reinforcing the sector's importance as a major component of Australia's service economy. Global comparisons further demonstrate that Australia is among the most internationalised education destinations, distinguishing it from most countries.

Overall, Australia's international education sector functions as a high-value, globally competitive, but structurally concentrated export industry, making it both economically powerful and strategically vulnerable in the face of external disruptions.

5. Reflection

This project demonstrates that effective data visualisation is not achieved through graphical complexity, but through alignment between data structure, analytical purpose, and interpretive framing. Clearer insights emerged when visualisations were simplified and selected based on their suitability for the underlying data rather than aesthetic considerations.

The process also highlighted that strong visual analysis depends fundamentally on robust data preparation. Working with multiple datasets required careful validation of types, time alignment, and category consistency — reinforcing that reliable insights are grounded in defensible data transformations rather than visual output alone.

Several limitations define the scope of these findings. The source-country analysis uses two time points rather than a continuous annual series, capturing broad directional change rather than year-by-year dynamics. The export analysis relies on aggregate revenue values, meaning changes in revenue per student may reflect pricing, course mix, or market composition rather than volume alone. The global comparison uses the latest available year per country, which varies slightly across sources, and the regression identifies association between enrolments and revenue without establishing causation. If extended further, the analysis would benefit from a complete longitudinal source-country dataset and additional macroeconomic indicators to allow deeper exploration of these relationships.

6. Bibliography

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7. AI and Software Declaration

This project was completed using R (version 4.x) and associated packages including tidyverse, ggplot2, readxl, forecast, cluster, and patchwork for all data wrangling, visualisation, statistical testing, and time-series modelling.

Generative AI tools were used as follows:

Tool: *Claude (Anthropic, claude.ai)*

Purpose and prompts used:

1. Code development and debugging — e.g. "Write R code to reshape this dataset from wide to long format for ggplot2 time-series visualisation", "Fix this ggplot2 error: [error message]"
2. Visualisation refinement — e.g. "Adjust this ggplot to use a diverging bar chart showing above and below trend values in green and red", "Add a LOESS smoother with 95% confidence band to this scatterplot"
3. ARIMA modelling — e.g. "Write R code using the forecast package to fit an ARIMA model to total enrolments and plot projections with 80% and 95% confidence intervals"
4. Wording refinement — e.g. "Improve the clarity of this paragraph while keeping the analytical interpretation unchanged"
5. APA referencing — e.g. "Format these dataset URLs as APA 7th edition bibliography entries"

All analytical decisions, interpretations, visualisation design choices, and final written content were made, reviewed, and validated by the student. The student takes full responsibility for the accuracy and integrity of all submitted content.

Note: a known rendering artifact in ggplot2 version 4.0.2 caused a spurious axis label to appear in Figures 1 and 2, this does not affect the underlying data or statistical analysis.